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COOKING APPLIANCE WITH CLIP

Abstract

A cooking appliance with a cooking area such as a stovetop or an oven can be utilized to prepare and cook food items. The cooking appliance includes a chassis forming a frame for the cooking appliance. A console couples to the chassis for operating the cooking appliance, such as including knobs or other controls for operating the stovetop or the oven. A clip removably couples the console to the chassis. The clip has a body including a spring finger for securing the console against the chassis between the spring finger and the chassis by a spring force.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates to a household appliance, such as a cooking appliance like a stove, an oven, or a combination thereof, and, to a clip for securing a console to a chassis of the cooking appliance.

BACKGROUND

[0002] Stovetops, such as gas stovetops, are utilized for cooking food or other activities that utilize heating materials with heightened temperatures. Stovetops that utilize combustible fuels, such as gas or charcoal, can utilize buttons, knobs, or other controls to operate burning of the combustible fuels, such as by controlling flow rates of a fluid or gas fuel to vary flame size and temperature output.

SUMMARY

[0003] In one aspect, the disclosure relates to a cooking appliance with a cooking area, the cooking appliance comprising: a chassis; a console coupled to the chassis for operating the cooking appliance; and a clip coupling the console to the chassis, the clip having a body including a spring finger; wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.

[0004] In another aspect, the disclosure relates to a clip for coupling a console to a chassis for a cooking appliance, the clip comprising: a body; a spring finger portion including a spring finger; and an extension spacing the spring finger portion from a remainder of the body; wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] In the drawings:

[0006] FIG. 1 depicts a cooking appliance with a cooktop and a console, in accordance with an aspect of the present disclosure.

[0007] FIG. 2 depicts the cooking appliance of FIG. 1 having the cooktop and the console exploded from the cooking appliance, exposing a chassis including an attachment wall for attaching to the console, in accordance with an aspect of the present disclosure.

[0008] FIG. 3 depicts a clip used to couple the console to the attachment wall of FIG. 2, in accordance with an aspect of the present disclosure.

[0009] FIG. 4 depicts a side view of the clip of FIG. 3, showing an angled orientation of a spring finger portion relative to an attachment portion, in accordance with an aspect of the present disclosure.

[0010] FIG. 5 depicts a side, sectional view of the clip of FIG. 3 coupled to the attachment wall of FIG. 2 by a fastener and having the console of FIG. 2 positioned above the clip to prepare for attachment of the console to the clip, in accordance with an aspect of the present disclosure.

[0011] FIG. 6 depicts another side, sectional view of the assembly of FIG. 5 having the console attached to the chassis at the attachment wall with the clip, in accordance with an aspect of the present disclosure.

[0012] FIG. 7 depicts a clip used to couple a console to a chassis for a cooking appliance, in accordance with an aspect of the present disclosure.

[0013] FIG. 8 depicts a side view of the clip of FIG. 7 showing an angled orientation of a spring finger portion relative to an attachment portion, in accordance with an aspect of the present disclosure.

[0014] FIG. 9 depicts a rear perspective view of the clip of FIG. 7, illustrating a tab and first and second wings utilized to ensure proper positioning of the clip, in accordance with an aspect of the

present disclosure.

[0015] FIG. **10** depicts an exploded, sectional view showing a chassis and a console exploded from the clip of FIG. **7**, in accordance with an aspect of the present disclosure.

[0016] FIG. **11** depicts a partially exploded, sectional view of the assembly of FIG. **10**, having the clip coupled to the chassis, in accordance with an aspect of the present disclosure.

[0017] FIG. **12** depicts a sectional view of the assembly of FIG. **11**, having the clip coupling the console to the chassis, in accordance with an aspect of the present disclosure.

[0018] FIG. **13** depicts a rear view of the assembly of FIG. **12**, illustrating a depression in the console attachment wall and how the first and second wings of FIG. **9** ensure proper positioning of the clip relative to the attachment wall at the depression, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

[0019] As may be appreciated, based on the disclosure, there exists a need in the art for an assembly facilitating attachment of a console to a chassis for a cooking appliance. The disclosure herein includes a clip that permits a user or manufacturer to easily align and attach a console unit to the chassis for the cooking appliance, as well as receive feedback that the proper attachment has been made. The force required to attach the console to the clip can be less than the force required to remove the console from the clip to facilitate installation, while preventing unintended removal of the console from the clip.

[0020] Referring to FIG. **1**, a cooking appliance **10** includes a cooktop **12** having a set of burners **14** defining an exterior cooking area **16**. The set of burners **14** in a non-limiting example can be gas burners, such as for burning synthetic gas, natural gas, propane, or other flammable liquids, gasses, or fuels in further non-limiting examples to cook or otherwise heat cooking items such as food items within the exterior cooking area **16**. While five burners are shown for the set of burners **14**, any number of burners are contemplated, including one or more.

[0021] A console **20** is positioned at a front of the cooktop **12**. A set of controllers **22**, depicted as knobs **24**, are utilized to control operation of the set of burners **14**, such as by turning the set of burners **14** on and off, as well as controlling flame size or heat produced with the set of burners **14**. In the illustrated example, each of the set of burners **14** has a corresponding one of the knobs **24**. A door **26** is positioned below the console **20** and includes a handle **28** to selectively open and close the door **26** for accessing an interior cooking area **30**, such as an oven. The set of controllers **22** can further include a control panel **32**, which can be provided on the console **20**. In a non-limiting example, the control panel **32** can be utilized to control operation of the interior cooking area **30**, such as by controlling the temperature of the interior cooking area **30**. While control of the cooking appliance **10** is illustrated by way of the knobs **24** and the control panel **32**, it is contemplated that any suitable system, feature, or controller can be incorporated into the console **20** to operate the cooking appliance **10**.

[0022] Referring to FIG. **2**, the cooktop **12** and the console **20** are exploded from the cooking appliance **10** to expose a chassis **40**. The control panel **32** is coupled to the chassis **40**, exposing an opening **34** in the console **20** when the console **20** is removed from the cooking appliance **10**. In an alternate, non-limiting example, it is contemplated that the control panel **32** is dedicated to the console **20** and attaches to the chassis **40** via the console **20**.

[0023] An attachment wall **42** couples to or forms a portion of the chassis **40**. To couple the console **20** to the cooking appliance **10**, the console **20** couples to the chassis **40** at the attachment wall **42**. The attachment wall **42** can be hidden by the cooktop **12** and the console **20** when the cooking appliance is fully assembled (see FIG. **1**).

[0024] FIG. **3** shows a clip **50** that can be utilized for coupling a console to a chassis for a cooking appliance, such as for coupling the console **20** to the attachment wall **42** for the cooking appliance **10** of FIG. **2**, for example. In a non-limiting example, the clip **50** can be for an electric cooking appliance, while a gas or other appliance or fuel type is contemplated. The clip **50** includes a body

52 having a spring finger portion **54** and a chassis attachment portion **56**. The spring finger portion **54** can be used to secure the clip **50** to the console **20** and the chassis attachment portion **56** can be used to secure the clip **50** to the chassis **40** at attachment wall **42** (see FIG. 2).

[0025] The spring finger portion **54** extends from the chassis attachment portion **56** at an extension portion **58**. An opening **60** is provided at least partially within the extension portion **58**, separating the extension portion **58** into a first extension **62** and a second extension **64**. The first and second extensions **62**, **64** can permit flexing or bending of the spring finger portion **54** relative to the chassis attachment portion **56** when acted upon by a large enough force. The size, material, thickness, and spacing among the first and second extensions **62**, **64** can define a spring force for flexion or bending of the spring finger portion **54**. A protuberance **66** can be provided in or on each of the first and second extensions **62**, **64** to reinforce the first and second extensions **62**, **64** against weakening when bending relative to the chassis attachment portion **56**. A spring finger **68** extends from the extension portion **58** opposite of the chassis attachment portion **56**. The spring finger **68** includes an exterior surface **70** and a contact surface **72** opposite the exterior surface **70**. The spring finger **68** includes a common curved or bent shape extending along a length (L).

[0026] The chassis attachment portion **56** extends between a first side **80** and a second side **82**, having a greater width than that of the spring finger portion **54**. A first alignment tab **84** extends from the chassis attachment portion **56** at the first side **80** and a second alignment tab **86** extends from the chassis attachment portion **56** at the second side **82**. A stop **88** extends from the chassis attachment portion **56** toward the spring finger portion **54** at the opening **60** and can be sized to fit within the opening **60**. A fastener opening **90** extends through the chassis attachment portion **56** and can be positioned centrally between the first side **80** and the second side **82**.

[0027] FIG. 4 illustrates the clip **50** from a side view, better illustrating the orientation of the spring finger portion **54** relative to the chassis attachment portion **56**. The extension portion **58** of the spring finger portion **54** is arranged at an angle **92** relative to the chassis attachment portion **56**. The angle **92** can be greater than 1-degree) (1° and less than or equal to 25-degrees) (25° in a non-limiting example, while any angle between 1-degree) (1° and 90-degrees) (90° is contemplated, such that the spring finger portion **54** is oriented at a non-zero angle relative to the chassis attachment portion **56**. In another non-limiting example, the angle can be 21-degrees) (21° . The spring finger **68** includes a concave curvature along the exterior surface **70** and a convex curvature along the contact surface **72**, further defining an upper contact surface **74** and a lower contact surface **76** resultant of the curvature of the spring finger **68**. The first and second alignment tabs **84**, **86** can be bent at a 90-degree angle relative to the remainder of the chassis attachment portion **56**. The stop **88** can be aligned with the chassis attachment portion **56**.

[0028] Referring to FIG. 5 the clip **50** provides for attaching the console **20** to the chassis **40** at the attachment wall **42**. Initially, the clip **50** can be coupled to the attachment wall **42**. A fastener hole **96** can be provided in the attachment wall **42** and is sized similar to the fastener opening **90** within the clip **50**. A fastener **98** couples the clip **50** to the attachment wall **42** through the fastener opening **90** at the fastener hole **96**.

[0029] Slots **100** (shown in broken line) extend through the attachment wall **42**. The first and second alignment tabs **84**, **86** can insert into the slots **100** during attachment of the clip **50** to the attachment wall **42**. Insertion of the first and second alignment tabs **84**, **86** into the slots **100** ensures proper alignment and positioning of the clip **50** relative to the attachment wall **42**. Furthermore, inserting the first and second alignment tabs **84**, **86** into the slots **100** prevents unintended rotation of the clip **50** when securing the clip **50** to the attachment wall **42**. For example, the fastener **98** may be a threaded fastener which can otherwise cause an unintended rotation of a clip relative to the attachment wall **42** without having the benefit of the first and second alignment tabs **84**, **86** inserted into the slots **100**.

[0030] The angled orientation of the spring finger portion **54** relative to the chassis attachment portion **56** results in a cavity **102** arranged between the spring finger **68** and the attachment wall **42**

when the clip **50** couples to the attachment wall **42**. The console **20** can include a console wall **104** with a protrusion **106**. The protrusion **106** can be sized to fit within the cavity **102**. The protrusion **106** can have a curved cross-sectional shape, similar to that of the spring finger **68**.

[0031] Referring to FIG. **6**, attachment of the console **20** to the attachment wall **42** is achieved by inserting the console wall **104** into the cavity **102**. The protrusion **106** is pushed down against the spring finger **68**, causing the spring finger **68** to bend and move away from the attachment wall **42** to permit room for the protrusion **106** to pass into the cavity **102**. More specifically, the protrusion **106** is pushed against the upper contact surface **74** to cause the spring finger **68** to bend away from the attachment wall **42** until enough room is provided for the protrusion **106** to pass into the cavity **102**. The console wall **104** can be fully inserted into the cavity **102** when it abuts the stop **88**. The elongated shape for the spring finger **68** defined by the length (L) (FIG. **3**) can form a lap joint with the protrusion **106**, among the parallel surfaces in contact with one another. Such a lap joint prevents a rotational offset between the spring finger **68** and the protrusion **106**.

[0032] Once the protrusion **106** is inserted into the cavity **102**, the spring finger **68** returns toward the attachment wall **42** utilizing a spring force created by the angled orientation of the spring finger portion **54** and by bending of the spring finger portion away from the attachment wall **42**. The spring force generated by the spring finger **68** creates a pressure against the console wall **104** between the clip **50** and the attachment wall **42** that secures the console wall **104** within the cavity **102** at the protrusion **106**. The elongated shape of the spring finger **68** having the length (L) (see FIG. **3**) can be complementary with an elongated shape for the protrusion **106**. The elongated shape of the spring finger **68** and a similar elongated shape for the protrusion **106** can prevent rotational misalignment between the spring finger **68** and the protrusion **106**, which can resist or prevent rotation of the console **20** relative to the clip **50**. In this way, proper alignment between the clip **50** and the console **20** is achieved and misalignment is prevented.

[0033] When the console **20** is properly inserted at the clip **50**, the spring finger **68** can generate an audible 'click' or other noise to indicate to the installer or user that attachment has been achieved. Such a click or noise can be generated by the spring finger **68** contacting the console **20** due to the spring force returning the spring finger **68** toward its original position after insertion of the protrusion **106** into the cavity **102**. Additionally, vibrations generated by the spring finger **68** can provide haptic feedback to the user to indicate that proper installation has occurred.

[0034] In a non-limiting example, spring finger **68** can be shaped and sized such that the force required for insertion of the protrusion **106** into the cavity **102** is greater than the force required for removal of the protrusion **106** from the cavity **102**. This can be achieved with the particular sizes and shapes for the spring finger **68** and the protrusion **106**. For example, the upper contact surface **74** can be arranged at a lesser angle than that of the lower contact surface **76**, relative to the console wall **104**, such that a greater force is required to displace the spring finger **68** when pressing against the upper contact surface **74** than when pulling against the lower contact surface **76**. In this way, a lesser force is required to attach the console **20** to the clip **50** than that required to remove the console **20** from the clip **50**. In another non-limiting example, the shape or size for the protrusion **106** can determine the amount of force required to attach or remove the console **20** from the clip **50**. For example, a lower surface **108** for the protrusion **106** can have a lesser angle of incline than an upper surface **110**, thereby requiring less force to insert the protrusion **106** into the cavity **102** than to remove the protrusion **106** from the cavity **102**. In a non-limiting example, the force required to attach the console wall **104** to the clip **50** can be 56.87 Newtons (N) and the force required to remove the console wall **104** from the clip **50** can be 65 N. In another non-limiting example, the force required to attach the console wall **104** to the clip **50** can be at least 3 N less than the force required to remove the console wall **104** from the clip **50**.

[0035] Referring to FIG. **7**, a clip **200** is provided that can be used to couple a console to a chassis for a cooking appliance, such as the console **20** and chassis **40** of FIGS. **1-2**, for example. In a non-limiting example, the clip **200** can be for a gas cooking appliance, while electric or other fuel types

are contemplated. The clip **200** includes a spring finger portion **202**, a center wall **204**, and a chassis attachment portion **206**, with the center wall **204** separating the spring finger portion **202** from the chassis attachment portion **206**. The spring finger portion **202** and the chassis attachment portion **206** each connect to the center wall **204** at a curved junction **208**, which can have a 180-degree turn between connected portions.

[0036] The spring finger portion **202** includes an extension portion **210** extending from the curved junction **208**. An opening **212** is provided at least partially within the extension portion **210**, defining a first extension **214** and a second extension **216**. A protuberance **218** can be provided on one or both of the first and second extensions **214**, **216**. A spring finger **220** extends from the extension portion **210** opposite of the curved junction **208**. The spring finger **220** includes a curved or bent shape similar to that discussed with respect to the spring finger **68** of FIGS. **3** and **4**. A stop **224** extends from the curved junction **208** toward the opening **212**. The chassis attachment portion **206** includes a first wing **230** and a second wing **232** and terminates at a lip **234** opposite of the curved junction **208**.

[0037] Referring to FIG. **8**, the stop **224** extends from the curved junction **208** parallel to the center wall **204**, while the spring finger portion **202** is arranged at an angle **236** offset from parallel to the center wall **204** at the extension portion **210**. The angle **236** can be greater than or equal to 1-degree) (1° and less than or equal to 25-degrees) (25° , while any angle greater than or equal to 1-degree) (1° and less than or equal to 90-degrees) (90° is contemplated. In one example, the angle **236** can be 21-degrees) (21° . The angled arrangement of the extension portion **210** defines a cavity **238** between the extension portion **210**, the spring finger **220**, and the center wall **204**. A tab **240** extends at an angle **248** from the chassis attachment portion **206** toward the center wall **204**. A first attachment region **244** can be defined as the area between the center wall **204** and the chassis attachment portion **206** and a second attachment region **246** can be defined as the area between the center wall **204** and the spring finger portion **202**.

[0038] FIG. **9** shows a rear view of the clip **200** with the tab **240** is at least partially positioned within and extending from an extension opening **242** formed in the chassis attachment portion **206**. The lip **234** extends between the first wing **230** and the second wing **232** and extends in a direction away from the center wall **204**.

[0039] Referring to FIG. **10**, a schematic section view illustrates the clip **200** separated from a console **250** and an attachment wall **252** or chassis. To attach the console **250** to the attachment wall **252** utilizing the clip **200**, the attachment wall **252** is inserted into the first attachment region **244** and the console **250** is inserted into the second attachment region **246**. The attachment wall **252** can further include a depression **254**, such that at least a portion of the attachment wall **252** forms a valley or basin that is unaligned with the remainder of the attachment wall **252**.

[0040] Referring to FIG. **11**, the clip **200** can attach to the attachment wall **252** at the depression **254** by inserting the attachment wall **252** into the first attachment region **244**, or by placing the clip **200** onto the attachment wall **252** and positioning the attachment wall **252** into the first attachment region **244**. Insertion of the attachment wall **252** into the first attachment region **244** depresses the tab **240** to permit the attachment wall **252** to extend to the curved junction **208** and fully insert into the first attachment region **244**. A tab opening **256** is provided in the attachment wall **252** to receive the tab **240** when the attachment wall **252** has been fully inserted into the first attachment region **244**. When the tab **240** inserts into the tab opening **256**, an audible “click” or other audible feedback is generated, instructing the installer or user that a proper attachment has occurred.

Additionally, the snap of the tab **240** inserting into the tab opening **256** provides haptic feedback to the user to inform the user that proper attachment has occurred. The tab **240** secures the clip **200** to the attachment wall **252** at the tab opening **256**, where the tab **240** prevents removal of the clip **200** from the first attachment region **244**. Removal of the clip **200** can be achieved by pushing against a bottom **258** of the lip **234**. More specifically, pushing the lip **234** can bend the chassis attachment portion **206** of the clip **200**, removing the tab **240** from the tab opening **256**, permitting removal of

the clip **200** from the attachment wall **252** at the first attachment region **244**. After attachment, at least a portion of the center wall **204** can position within the depression **254**.

[0041] Referring to FIG. **12**, the clip **200** can attach to the console **250** by inserting the console **250** into the second attachment region **246**. The console **250** does not extend into the depression **254** due to the length of the console **250** being longer than that of the depression **254**, thereby preventing the console **250** from entering the depression **254**. During attachment, when the console **250** is moving into the second attachment region **246**, a protrusion **260** on the console **250** forces the spring finger **220** in a direction away from the center wall **204**, providing room for the protrusion **260** to pass by the spring finger **220** and insert into the cavity **238**. The spring finger **220** can form a lap joint with the protrusion **260**, which can prevent a rotational offset among the spring finger **220** and the protrusion. After the protrusion **260** moves into the cavity **238**, the spring finger **220** returns toward its original position because of a spring force generated by the spring finger **220** bending away from the attachment wall **252**. The protrusion **260** secures the console **250** within the second attachment region **246** at the spring finger **220**. The spring force imparted by the spring finger **220** against the console **250** presses the console **250** against the attachment wall **252** to secure the console **250**. When the protrusion **260** is secured within the cavity **238**, the console **250** can rest against the stop **224**.

[0042] When the console **250** is properly inserted, the spring finger **220** can generate an audible 'click' or other noise to indicate to the installer or user that attachment has been achieved. Such a click or noise can be generated by the spring finger **220** contacting the console **250** due to the spring force returning the spring finger **220** toward its original position. Additionally, vibrations generated by the spring finger **220** can provide haptic feedback to the user to indicate that proper installation has occurred.

[0043] Removal of the console **250** from the clip **200** can be achieved by depressing the spring finger **220** away from the console **250**, permitting the protrusion **260** to exit the cavity **238**. The console **250** can then be removed from the second attachment region **246**.

[0044] Referring to FIG. **13**, the depression **254** is more-readily appreciable in the attachment wall **252**, and the depression **254** extends out of the page from the rear-view of FIG. **13**. As can be appreciated, the span for the first and second wings **230**, **232** is greater than a width (W) for the depression **254**, thereby preventing improper insertion of the clip **200**. Due to the span of the first and second wings **230**, **232**, the clip **200** is only insertable against a rear surface **262** of the attachment wall **252**, thereby ensuring proper positioning of the clip **200** at the attachment wall **252**. If a user or installer attempts to attach the clip **200** with the first and second wings **230**, **232** extending into the depression **254**, the span of the first and second wings **230**, **232** being greater than the width (W) of the depression **254** will prevent insertion of the attachment wall **252** into the first attachment region **244**, as the first and second wings **230**, **232** will contact the depression **254**. Such an interference is shown in broken line in FIG. **13**, preventing further insertion into the depression **254** by the first and second wings **230**, **232** against the attachment wall **252**. The tab **240** is not permitted to reach the tab opening **256**. In this way, proper orientation and attachment of the clip **200** relative to the attachment wall **252** is ensured.

[0045] Benefits can be realized by use of the clip **50**, **200** for attaching a console **20**, **250** to a chassis **40** or attachment wall **42**, **252** of the chassis **40**. The clip **50**, **200** as described herein provides for improvements for attachment of a console **20**, **250** to a chassis **40**. The clip **50**, **200** provides feedback to the user or installer, such as audible or haptic feedback, to let the user or installer know that a proper attachment has been achieved. The insertion force required to install the console **20**, **250** to the clip **50**, **200** can be less than the force required to remove the console **20**, **250** from the clip **50**, **200** which facilitates installation, while reducing the possibility of unintended detachment of the console **20**, **250** at the clip **50**, **200**. The shape for the spring finger **68**, **220** on the clip **50**, **200** forms a lap joint with the protrusion **106**, **260** on the console **20**, **250** which prevents a rotational offset for the clip **50**, **200** relative to the console **20**, **250**. Similarly, the first

and second alignment tabs **84, 86** ensure proper installation position, and prevent rotational misalignment between the clip **50** and the attachment wall **42** of the chassis **40**. Furthermore, the first and second wings **230, 232** prevent incorrect orientation of the clip **200** relative to the attachment wall **252**. These features ensure proper positioning and alignment of the clip **50, 200**, and therefore proper alignment and positioning of the console **20, 250** relative to the chassis **40**. [0046] Further benefits can include a positive removable lock for the clip at both the console **20, 250** and the chassis **40** or attachment wall **42, 252**. Specifically, the tab **240** inserting into the tab opening **256** on the attachment wall **252** ensures proper positioning of the clip **200**, as well as preventing unintended separation of the clip **200** from the attachment wall **252**. The lip **234** permits positive removal of the clip **200** from the attachment wall **252**.

[0047] Further still, the design of the clip **50, 200** improves assembly and installation, reducing manufacture and finishing costs and time, as well as improving serviceability of the cooking appliance **10**.

[0048] Many other possible embodiments and configurations in addition to that shown in the above figures are contemplated by the present disclosure. To the extent not already described, the different features and structures of the various embodiments can be used in combination with each other. That one feature cannot be illustrated in all the embodiments is not meant to be construed that it cannot be but is done for brevity of description. Thus, the various features of the different embodiments can be mixed and matched to form new embodiments, whether or not the new embodiments are expressly described. Moreover, while “a set of” various elements have been described, it will be understood that “a set” can include any number of the respective elements, including only one element. Combinations or permutations of features described herein are covered by this disclosure.

[0049] This written description uses examples to disclose embodiments of the disclosure, including the best mode, and to enable any person skilled in the art to practice embodiments of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0050] Further aspects are provided by the subject matter of the following clauses:

[0051] A cooking appliance with a cooking area, the cooking appliance comprising: a chassis; a console coupled to the chassis for operating the cooking appliance; and a clip coupling the console to the chassis, the clip having a body including a spring finger; wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.

[0052] The cooking appliance of any preceding clause, wherein the clip further comprises a chassis attachment portion for coupling the clip to the chassis.

[0053] The cooking appliance of any preceding clause, wherein the spring finger is offset from the chassis attachment portion by an angle.

[0054] The cooking appliance of any preceding clause, wherein the angle is greater than or equal to 1-degree and less than or equal to 25-degrees.

[0055] The cooking appliance of any preceding clause, further comprising a tab provided on the chassis attachment portion.

[0056] The cooking appliance of any preceding clause, wherein the chassis further comprises a tab opening for receiving the tab when the clip is attached to the chassis.

[0057] The cooking appliance of any preceding clause, wherein the chassis attachment portion further comprises a lip.

[0058] The cooking appliance of any preceding clause, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger.

[0059] The cooking appliance of any preceding clause, wherein the clip further comprises a first attachment region defined between the center wall and the chassis attachment portion, and wherein the chassis is received within the first attachment region when the clip is coupled to the chassis.

[0060] The cooking appliance of any preceding clause, wherein the clip further comprises a second attachment region defined between the center wall and the spring finger, and wherein the console is received within the second attachment region when the clip is coupled to the console.

[0061] The cooking appliance of any preceding clause, wherein the center wall meets each of the chassis attachment portion and the spring finger at a curved junction.

[0062] The cooking appliance of any preceding clause, wherein the spring finger is shaped such that a force required to attach the console to the clip is lesser than a force required to remove the console from the clip.

[0063] The cooking appliance of any preceding clause, wherein the console further comprises a console wall having a bump sized to insert at the spring finger.

[0064] The cooking appliance of any preceding clause, wherein the console further comprises a protrusion sized to be received by the spring finger.

[0065] The cooking appliance of any preceding clause, wherein the protrusion meets the spring finger at a lap joint.

[0066] The cooking appliance of any preceding clause, further comprising a pair of extensions spacing the spring finger from a remainder of the body.

[0067] The cooking appliance of any preceding clause, wherein the spring finger is flexible relative to the body at the pair of extensions to generate the spring force.

[0068] The cooking appliance of any preceding clause, further comprising a protuberance provided on each of the pair of extensions.

[0069] The cooking appliance of any preceding clause, further comprising an opening in the body between the pair of extensions.

[0070] The cooking appliance of any preceding clause, further comprising a stop extending from the body toward the opening.

[0071] The cooking appliance of any preceding clause, wherein the body further comprises a fastener opening for coupling the clip to the chassis with a fastener.

[0072] The cooking appliance of any preceding clause, wherein the chassis comprises a fastener hole for receiving the fastener from the fastener opening.

[0073] The cooking appliance of any preceding clause, further comprising an alignment tab extending perpendicular from the body.

[0074] The cooking appliance of any preceding clause, wherein the chassis further comprises a slot for receiving the alignment tab when the clip is coupled to the chassis.

[0075] The cooking appliance of any preceding clause, further comprising a pair of wings extending from the body to define a span among the pair of wings.

[0076] The cooking appliance of any preceding clause, wherein the chassis includes a depression having a width that is less than the span among the pair of wings.

[0077] A clip for coupling a console to a chassis for a cooking appliance, the clip comprising: a body; a spring finger portion including a spring finger; and an extension spacing the spring finger portion from a remainder of the body; wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.

[0078] The clip of any preceding clause, wherein the extension is one of a pair of extensions spacing the spring finger portion from the remainder of the body, defining an opening between the pair of extensions.

[0079] The clip of any preceding clause, wherein the spring finger portion is flexible relative to the body at the pair of extensions to generate the spring force.

[0080] The clip of any preceding clause, further comprising a protuberance provided on each of the pair of extensions.

[0081] The clip of any preceding clause, further comprising an opening in the body between the pair of extensions.

[0082] The clip of any preceding clause, further comprising a stop extending from the body toward the opening.

[0083] The clip of any preceding clause, further comprising an alignment tab extending perpendicular from the body.

[0084] The clip of any preceding clause, further comprising a pair of wings extending from the body to define a span among the pair of wings.

[0085] The clip of any preceding clause, wherein the clip further comprises a chassis attachment portion.

[0086] The clip of any preceding clause, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger portion.

[0087] The clip of any preceding clause, wherein the center wall meets each of the chassis attachment portion and the spring finger portion at a curved junction to define a first attachment region between the center wall and the spring finger portion and a second attachment region between the center wall and the chassis attachment portion.

[0088] The clip of any preceding clause. further comprising a tab provided on the chassis attachment portion.

[0089] The clip of any preceding clause. further comprising a lip extending from the chassis attachment portion.

Claims

1. A cooking appliance with a cooking area, the cooking appliance comprising: a chassis; a console coupled to the chassis for operating the cooking appliance; and a clip coupling the console to the chassis, the clip having a body including a spring finger; wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.
2. The cooking appliance of claim 1, wherein the clip further comprises a chassis attachment portion for coupling the clip to the chassis.
3. The cooking appliance of claim 2, wherein the spring finger is offset from the chassis attachment portion by an angle.
4. The cooking appliance of claim 3, wherein the angle is greater than or equal to 1-degree and less than or equal to 25-degrees.
5. The cooking appliance of claim 2, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger.
6. The cooking appliance of claim 5, wherein the clip further comprises a first attachment region defined between the center wall and the chassis attachment portion, and wherein the chassis is received within the first attachment region when the clip is coupled to the chassis.
7. The cooking appliance of claim 6, wherein the clip further comprises a second attachment region defined between the center wall and the spring finger, and wherein the console is received within the second attachment region when the clip is coupled to the console.
8. The cooking appliance of claim 1, wherein the spring finger is shaped such that a force required to attach the console to the clip is lesser than a force required to remove the console from the clip.
9. The cooking appliance of claim 1, wherein the console further comprises a protrusion sized to be received by the spring finger.
10. The cooking appliance of claim 9, wherein the protrusion meets the spring finger at a lap joint.
11. The cooking appliance of claim 1, further comprising a pair of extensions spacing the spring finger from a remainder of the body.
12. The cooking appliance of claim 1, further comprising an alignment tab extending perpendicular from the body.

- 13.** The cooking appliance of claim 12, wherein the chassis further comprises a slot receiving the alignment tab when the clip is coupled to the chassis.
- 14.** The cooking appliance of claim 1, further comprising a pair of wings extending from the body to define a span among the pair of wings.
- 15.** The cooking appliance of claim 14, wherein the chassis includes a depression having a width that is less than the span among the pair of wings.
- 16.** A clip for coupling a console to a chassis for a cooking appliance, the clip comprising: a body; a spring finger portion including a spring finger; and an extension spacing the spring finger portion from a remainder of the body; wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.
- 17.** The clip of claim 16, wherein the extension is one of a pair of extensions spacing the spring finger portion from the remainder of the body, defining an opening between the pair of extensions.
- 18.** The clip of claim 16, wherein the body further comprises a chassis attachment portion.
- 19.** The clip of claim 18, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger portion.
- 20.** The clip of claim 19, wherein the center wall meets each of the chassis attachment portion and the spring finger portion at a curved junction to define a first attachment region between the center wall and the spring finger portion and a second attachment region between the center wall and the chassis attachment portion.
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